Ebrahim Abdelghfar Ebrahim

Robotics Software Engineer | Autonomous Navigation, Motion Control, Simulation
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Summary

I'm a robotics software engineer who writes navigation and control software for mobile robots and autonomous vehicles, and builds the simulators that test it before it goes into the field. In almost three years I've delivered a ROS 2/Nav2 stack for a mobile robot, designed MPC for a four-wheel independent-steering platform, and built CARLA SIL and HIL test benches at Micropolis that cut on-site testing by 70%.

Most of my work is in C++ and Python on Linux. I like tooling that makes a team faster: Docker images that just work, CI/CD that catches problems early, and simulators people actually want to use. I'm finishing an M.Sc. in Mechatronics Engineering at Ain Shams University (June 2027), and alongside it I'm working on on-track tire identification and adaptive nonlinear MPC.

Experience

Robotics Simulation Engineer | Micropolis, Dubai, UAE · 04/2026 – Present
- Built the software-in-the-loop (SIL) pipeline in CARLA 0.9.16 and Unreal Engine 4.26, with realistic LiDAR, camera and GNSS models and vehicle dynamics. This cut on-site testing by 70%.
- Set up a hardware-in-the-loop (HIL) bench that feeds simulated sensor data to the vehicle in the formats its drivers expect, so the onboard stack runs against the simulator without changes.
- Made a GUI tool on top of CARLA ScenarioRunner so the team can build automated test scenarios visually rather than coding each one.
- Brought UAV-captured dense point clouds into CARLA through custom data pipelines to make the simulated environments more faithful.
- Adapted NVIDIA Cosmos-Drive-Dreams to generate synthetic training data from CARLA scenes, helping the perception stack close the sim-to-real gap.

Robotics Systems Engineer | Dakahlia Group, Cairo, Egypt · 12/2024 – 04/2026
- Delivered the ROS 2 / Nav2 navigation stack for the company's mobile robot, combining cuVSLAM visual odometry with RTAB-Map occupancy-grid mapping. Localisation error stayed within 0.17 m over a ~1 km run.
- Designed the Model Predictive Controller (MPC) for the four-wheel independent-steering robot.
- Set up SIL benchmarks in Gazebo and NVIDIA Isaac Sim, which cut hardware test cycles by 40%.
- Packaged the robot software in Docker, so setting up a new machine took 75% less time and every deployment came out the same.
- Wrote the platform's technical roadmap and development plan.

Robotics Systems Engineer | I-Robotic, Cairo, Egypt · 01/2024 – 11/2024
- Designed the software stack for the Robo-Cafe product and built a detailed CoppeliaSim simulator for it, which lowered development cost by about 30%.
- Connected industrial PLCs to high-level ROS controllers for factory-automation projects.
- Set up the company's Git workflow and CI/CD pipeline.

Selected Projects

On-Track System Identification & Adaptive Nonlinear MPC | Research project
- Built a pipeline that estimates Pacejka Magic Formula tire coefficients on the track, starting with no prior vehicle data.
- Wrote a supervisory controller: the car first drives on Pure Pursuit to gather excitation data, then switches to nonlinear MPC once the parameters are known.
- Drove stably on a low-friction surface ($\mu = 0.35$) at up to 100 km/h.

Formula Student AI Simulator | CARLA 0.9.16, Unreal Engine 4.26 | 07/2026 – 08/2026
- Built a simulator for Formula Student AI teams and researchers. Sensor placement, parameters and rates are all set in config, so teams can start testing right away.

Path-Tracking Controller Benchmark | Python, ROS 2, CARLA
- Implemented Pure Pursuit, Stanley and MPC path tracking and compared how each performs in CARLA through ROS 2.

Rotary Inverted Pendulum: LQR with Hardware-in-the-Loop | MATLAB/Simulink, Autodesk Inventor
- Designed the mechanism in Inventor, derived a state-space model from its nonlinear dynamics, and balanced it with LQR in Simulink, with the hardware in the loop.

Skills

- **Programming:** C++, Python, C, Lua
- **Robotics software:** ROS 2, ROS 1, Nav2, MoveIt 2, RTAB-Map, NVIDIA cuVSLAM
- **Localization & mapping:** SLAM, visual odometry, multi-sensor fusion, Extended Kalman Filter (EKF), A* path planning
- **Control:** Model Predictive Control (MPC, nonlinear MPC), LQR, Pure Pursuit, Stanley, PID, system identification, vehicle and tire dynamics
- **Perception:** YOLOv5, depth cameras, LiDAR, camera and GNSS sensor modelling, synthetic data generation (NVIDIA Cosmos)
- **Simulation & testing:** CARLA, ScenarioRunner, Unreal Engine, NVIDIA Isaac Sim, Gazebo (Classic, Harmonic), CoppeliaSim, Webots, software-in-the-loop (SIL), hardware-in-the-loop (HIL)
- **Engineering tools:** Linux, Git, Docker, CMake, CI/CD, CAN bus, TCP/UDP, PLC integration, MATLAB/Simulink, Autodesk Inventor, Blender

Education

M.Sc. Mechatronics Engineering | Ain Shams University, Cairo | Expected 06/2027

B.Sc. Mechatronics Engineering | Ain Shams University, Cairo | 09/2018 – 07/2023

- Thesis: an autonomous robot arm for non-destructive e-waste disassembly. Developed the approach-and-unscrew algorithm (98% success rate) and screw localisation with YOLOv5 and a depth camera.
- Built a 10-method SDK that simplifies MoveIt motion planning and frame transforms, plus a CoppeliaSim/ROS 1 digital twin for testing.

Teaching & Leadership

Autonomous Racing Team Coordinator | Autotronics Research Lab (ARL), Cairo · 06/2023 – 04/2025 (part-time)

- Built and Dockerised a Shell Eco-marathon Urban Concept simulator in Gazebo Harmonic and ROS 2, with sensor models and vehicle dynamics, that runs on low-spec laptops. Also set up CARLA environments for testing ADAS algorithms.
- Planned and taught a 3-month autonomous-systems workshop for 100 students and mentored teams entering autonomous-driving competitions.

Autonomous Systems Mentor | Egyptian Electric Vehicle Rally Competition, Cairo · 10/2023 – 08/2024 (part-time)

- Built the competition's milestone simulator in CoppeliaSim and ROS 1, used by the competing teams.
- Wrote the Pure Pursuit steering controller for the autonomous car that drove the opening parade lap.
- Designed and delivered autonomous-systems training for national teams.

Certification & Languages

- Robotics Software Engineer Nanodegree, Udacity. Capstone: home-service robot with A*, EKF and PID in C++/ROS, tested in Gazebo.
- Languages: Arabic (native), English